



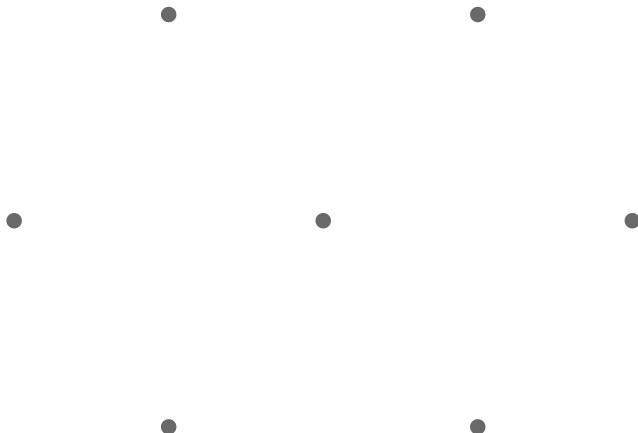
Technische
Universität
Braunschweig



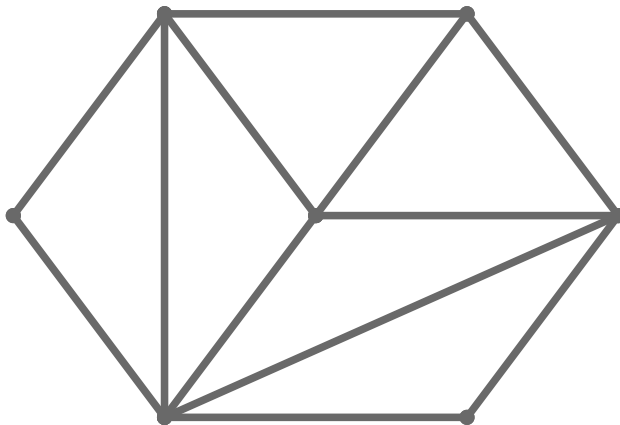
Algorithmische Methoden zu gradbeschränkten Triangulierungen

Fabian Alich, 4. September 2025

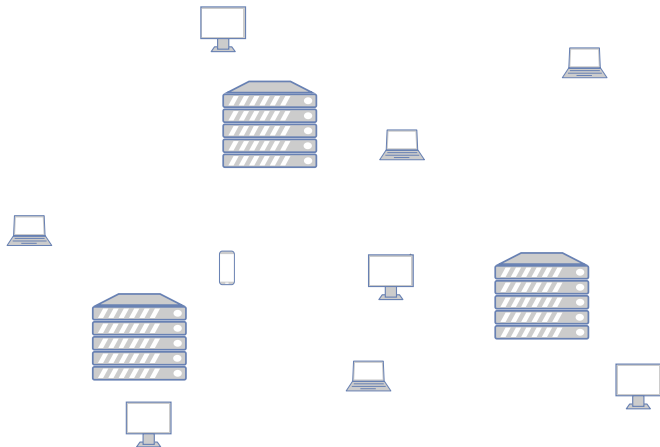
Triangulierung



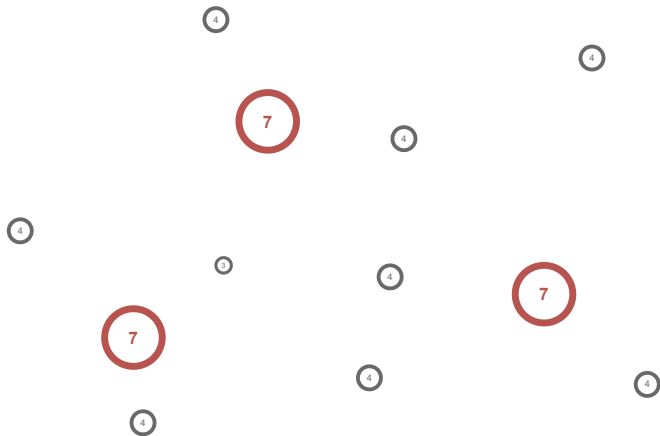
Triangulierung



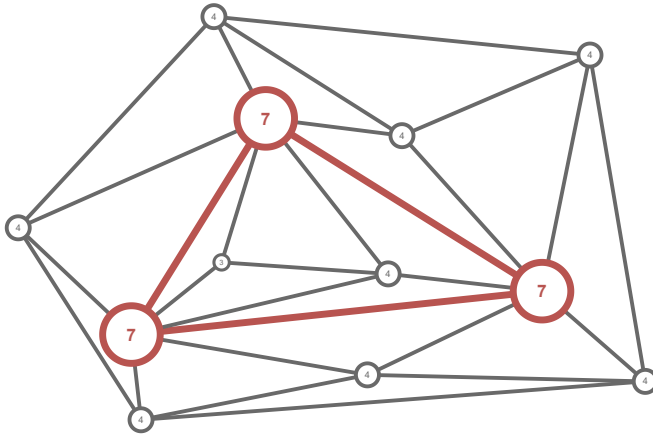
Praktische Relevanz



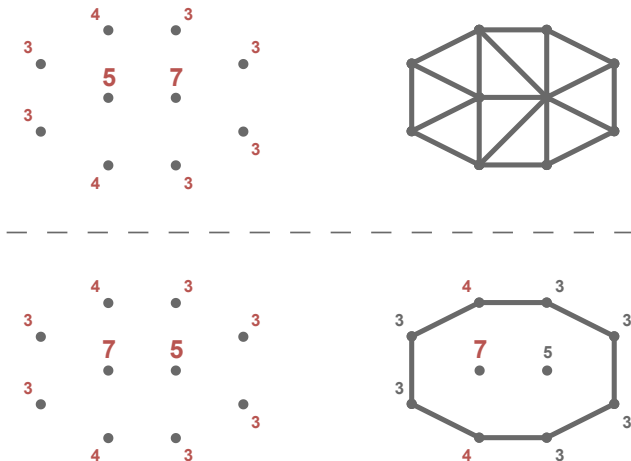
Praktische Relevanz



Praktische Relevanz



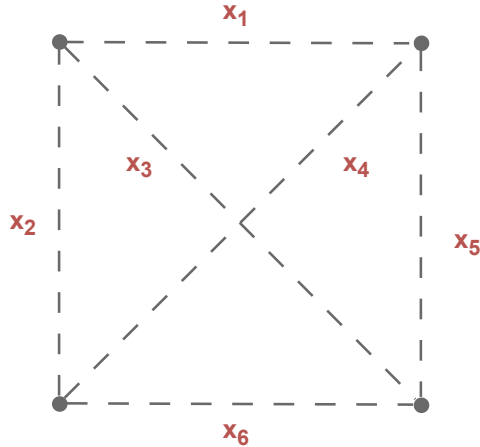
Das Problem



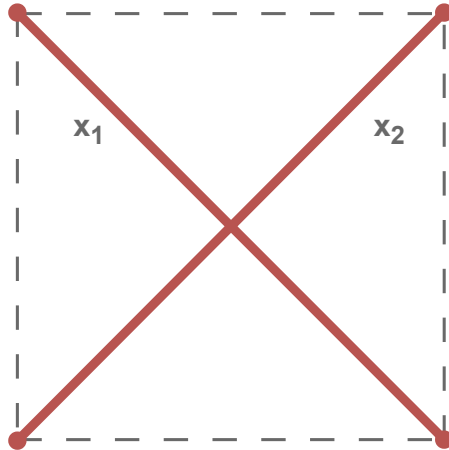
Teil I

Lösungsansätze

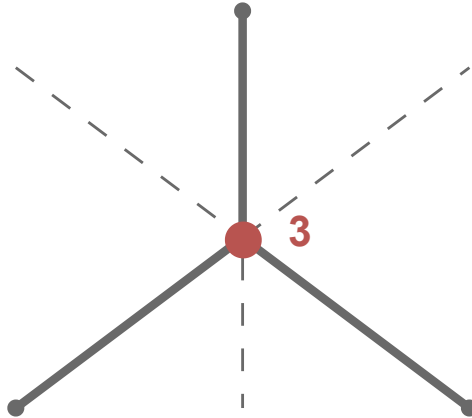
Kantenansatz



Überschneidungsbedingung



Gradbeschränkung (Entscheidungsproblem)

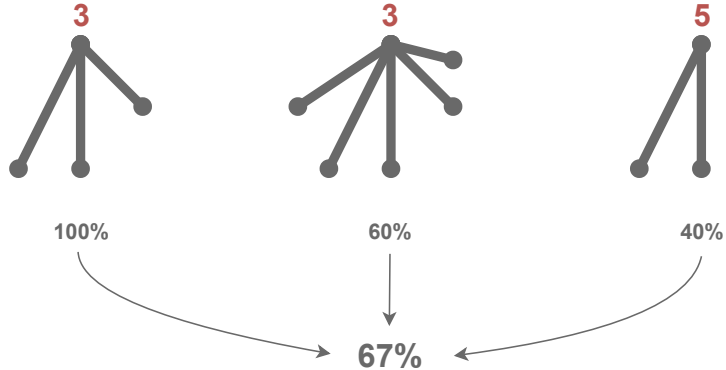


Maximale Kanten (Entscheidungsproblem)

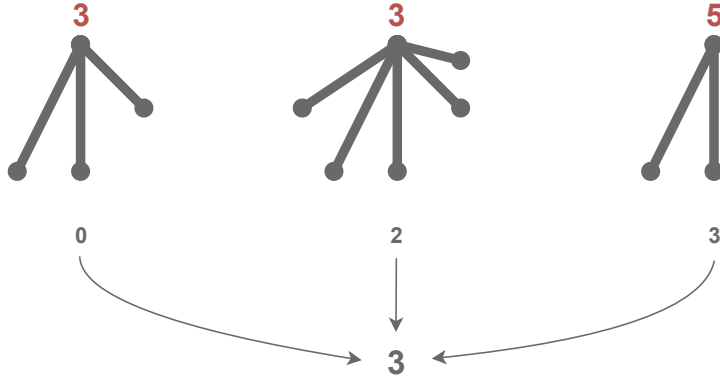
FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \stackrel{!}{=} \sum \text{Sollgrad}$$

Gradbeschränkung (Optimierungsproblem)

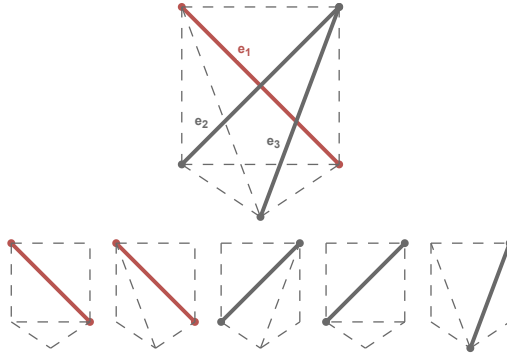


Gradbeschränkung (Optimierungsproblem)

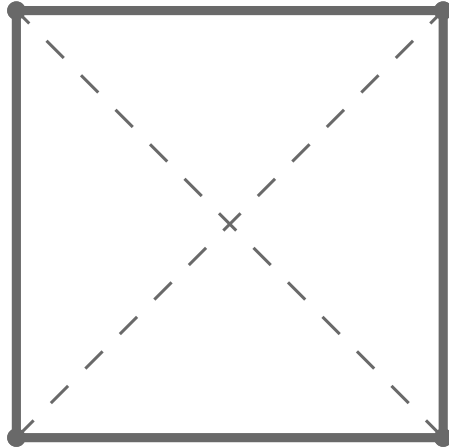


alternative Kantenbedingung

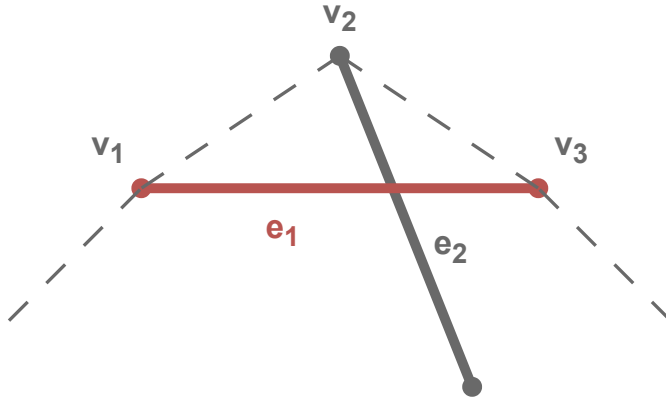
FA: sollte ich hier eine Refernz auf euer Paper machen?



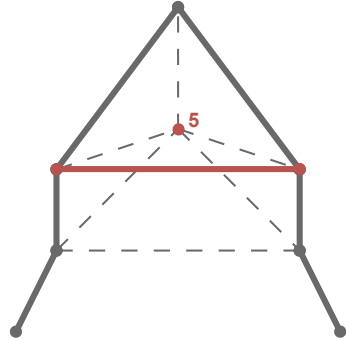
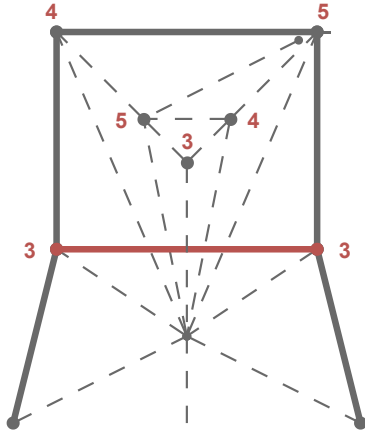
Hülle Fixieren



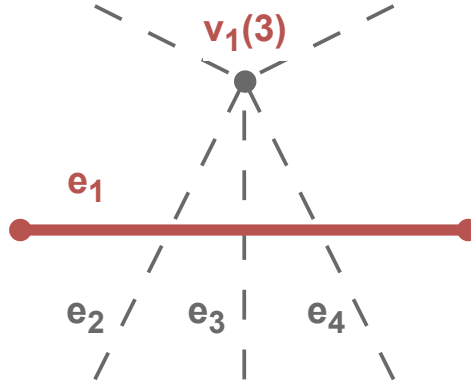
Kanten Fixieren / Ausschließen



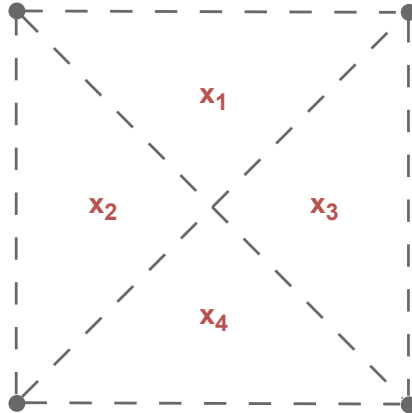
Kanten Ausschließen



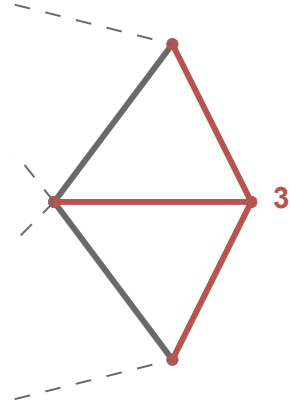
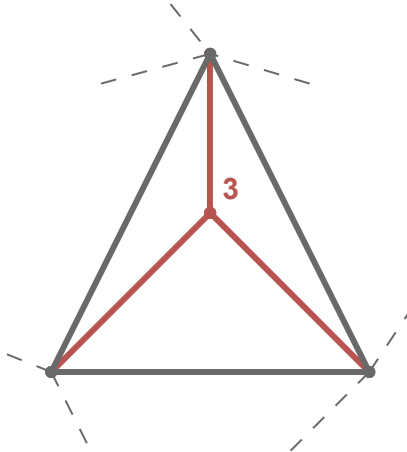
Kanten Ausschließen



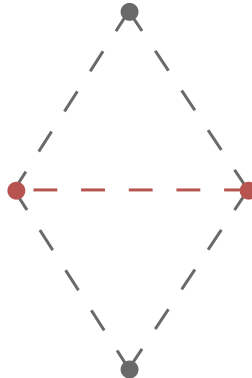
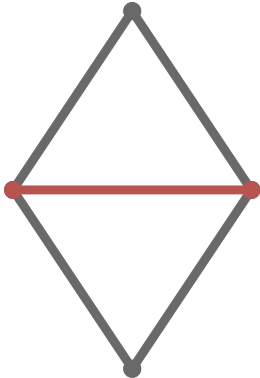
Dreiecksansatz



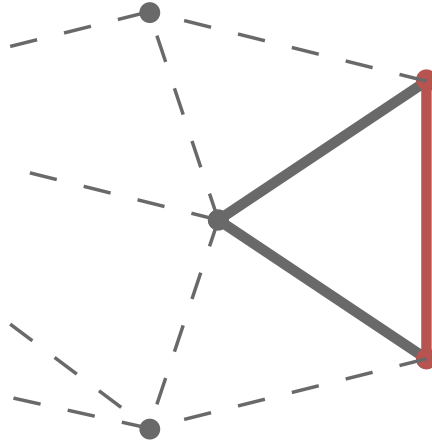
Gradbeschränkung



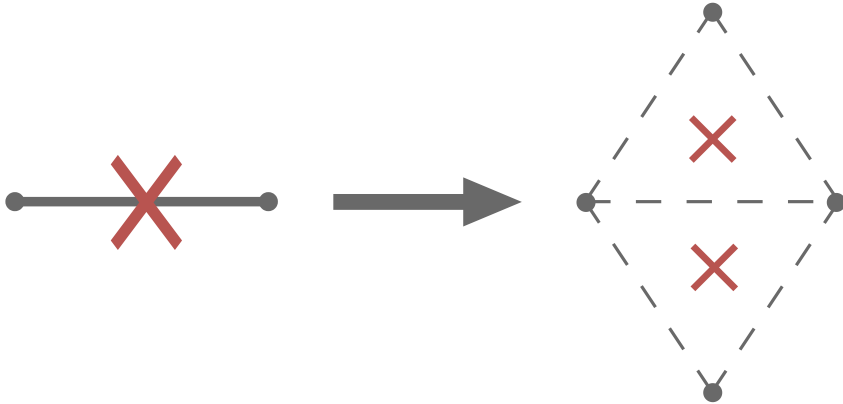
Überschneidungsbedingung



Überschneidungsbedingung



Dreiecks Ausschließen



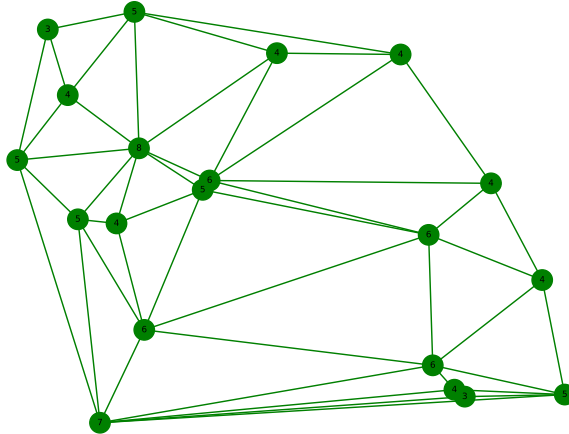
Subset

FA: ich habe jetzt subset in den Diagrammen drin, ist aber schlecht. Soll ich das dann hier erklären oder wie mache ich das am besten? Einfach nur auf die Arbeit verweisen?

Teil II

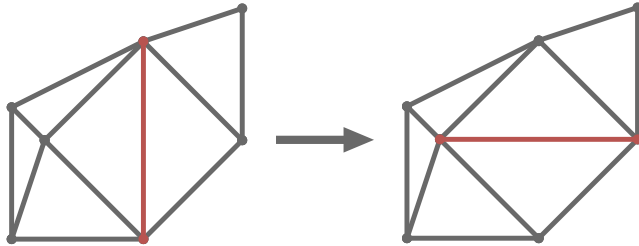
Instanzgenerierung

Delaunay

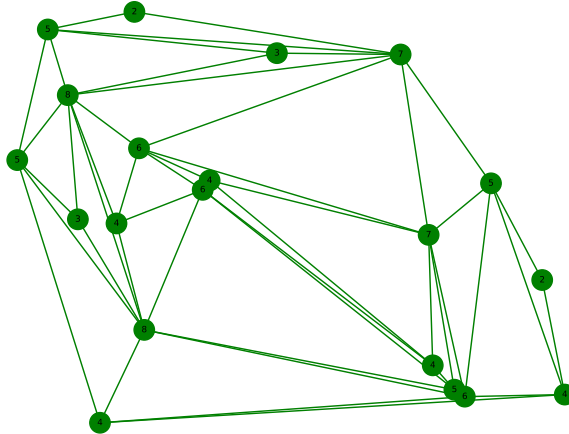


Flip

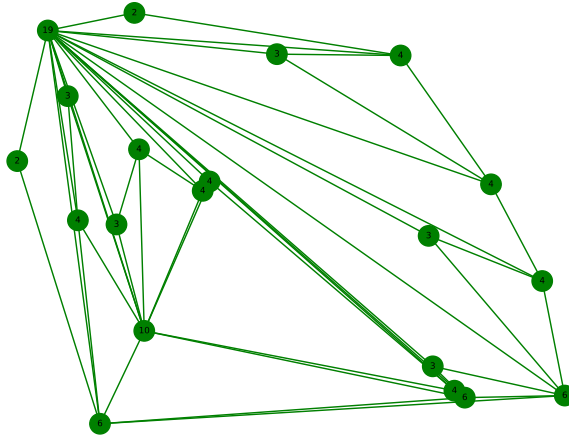
FA: Soll ich das als Animation



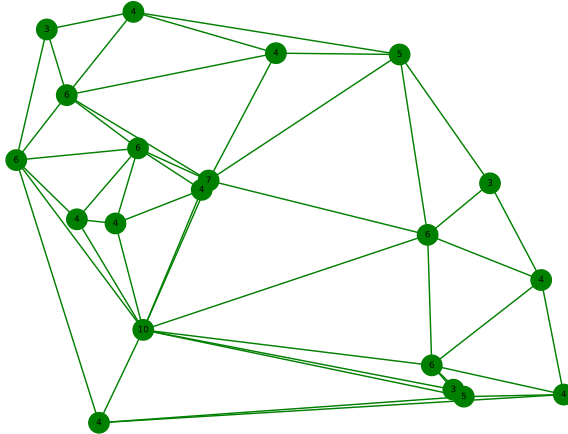
Delaunay-Flips



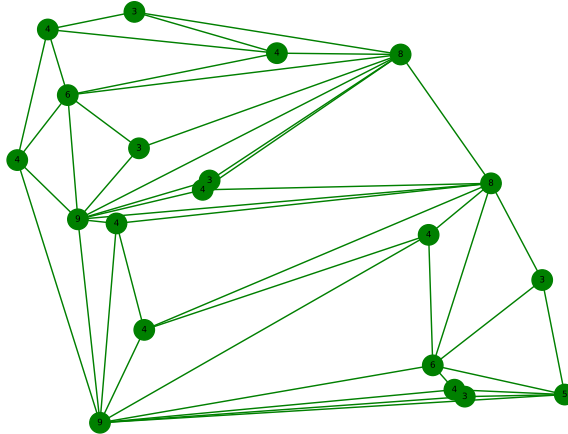
Iterativ



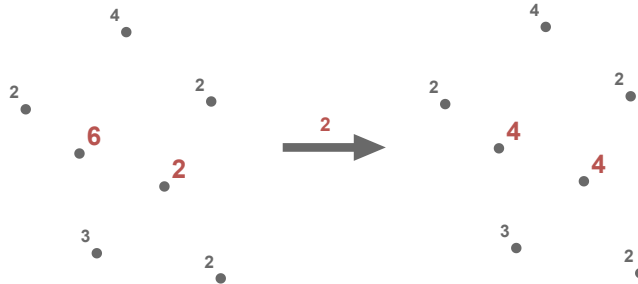
Greedy



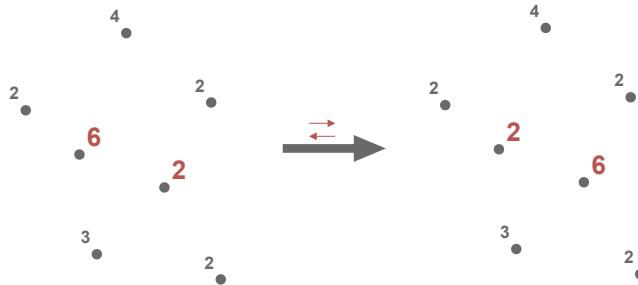
Random



Move



Change



Teil III

Auswertung

Solver



Gurobi



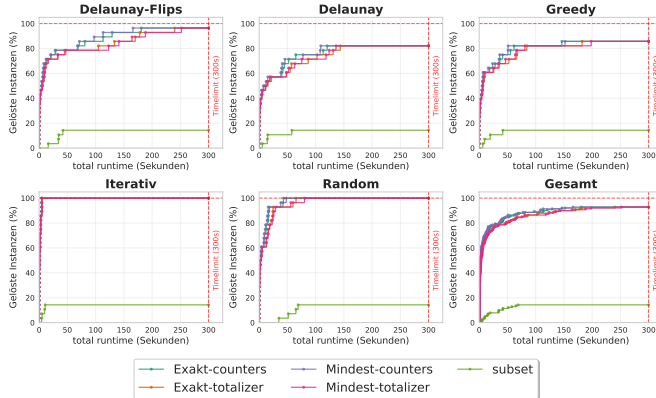
OR-Tools



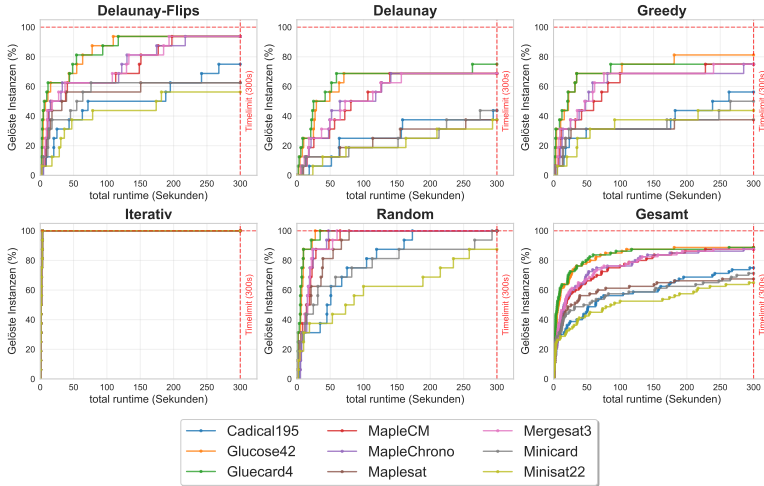
PySAT

Grad Codierung

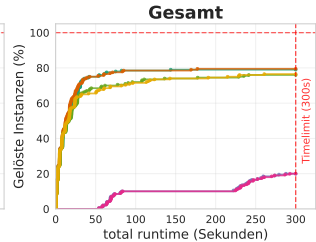
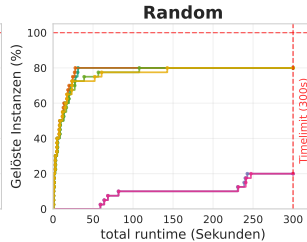
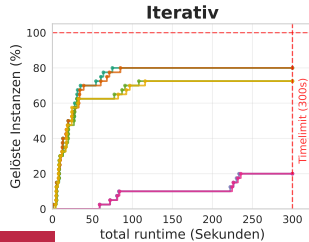
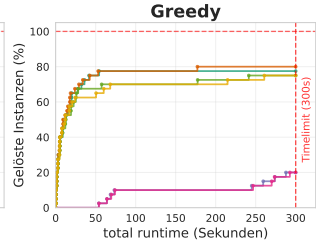
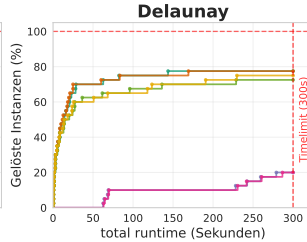
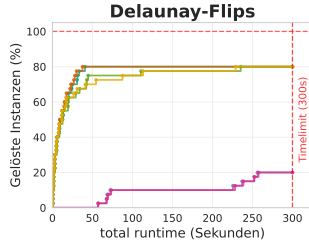
FA: soll ich hier immer sagen welche Knotengrößen?



PySAT Solver



Dreiecke Bedingungen



(tri) Kanten ausschließen
(tri) normal

PySAT (tri) Kanten ausschließen

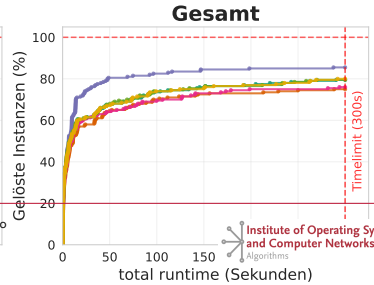
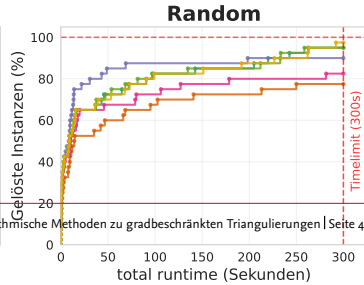
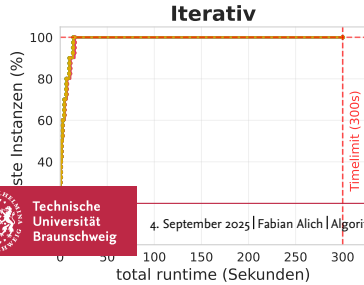
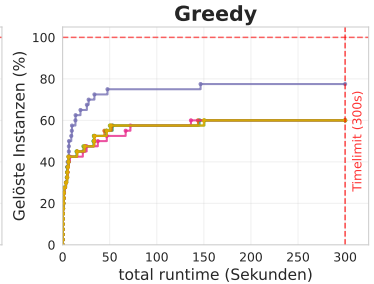
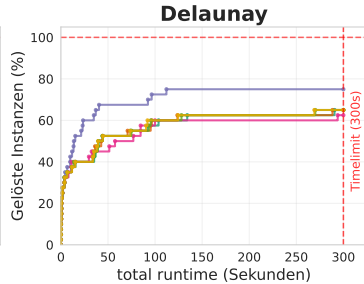
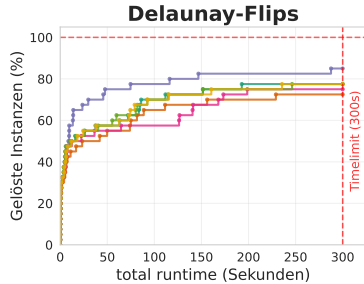
PySAT (tri) normal

Gurobi (tri) Kanten ausschließen

Gurobi (tri) normal

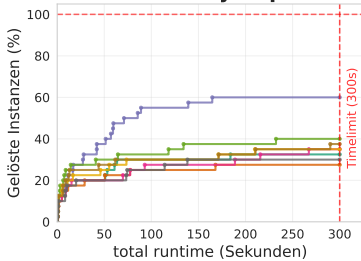
Institute of Operating Systems and Computer Networks Algorithms

PySAT Bedingungen

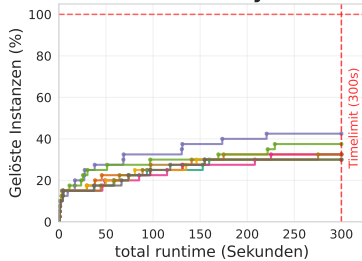


Gurobi Bedingungen

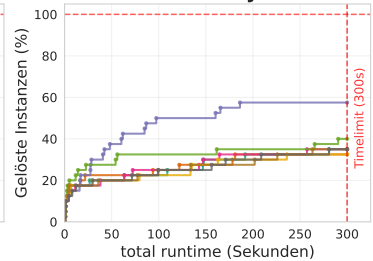
Delaunay-Flips



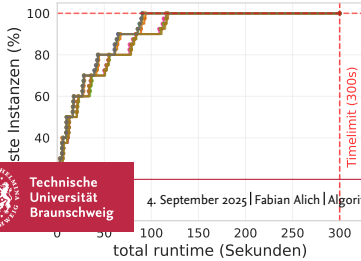
Delaunay



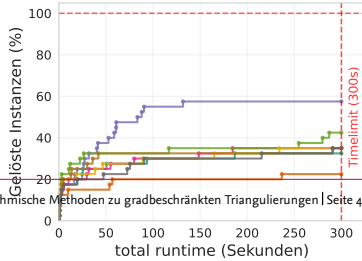
Greedy



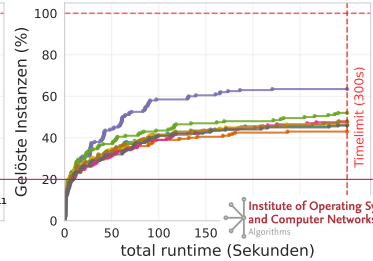
Iterativ



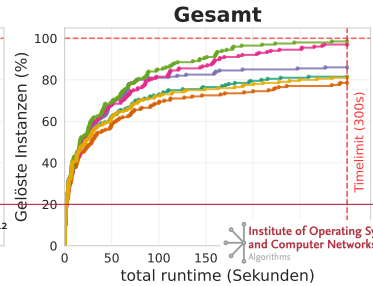
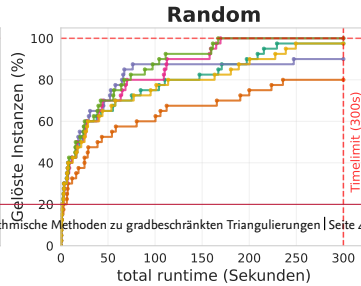
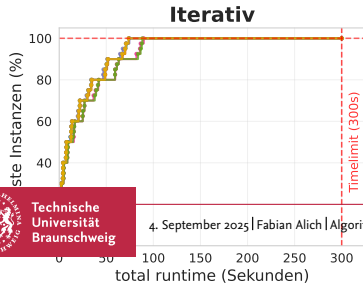
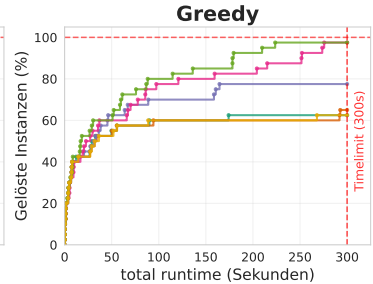
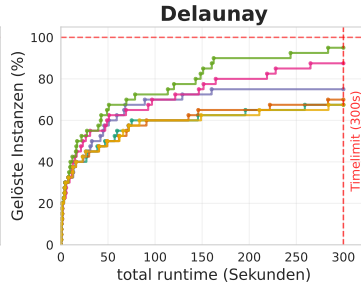
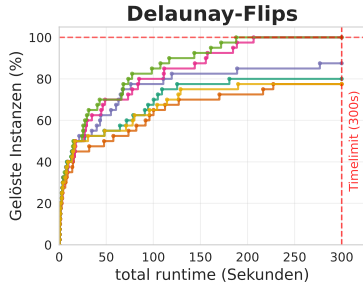
Random



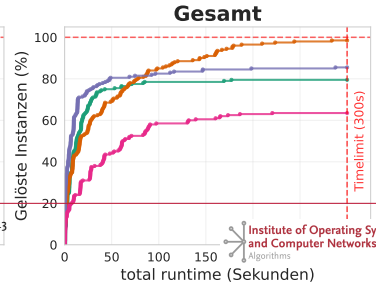
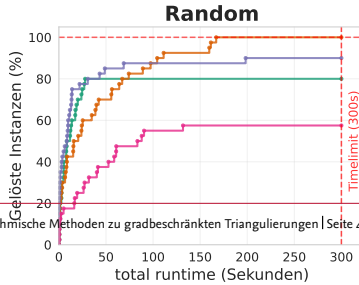
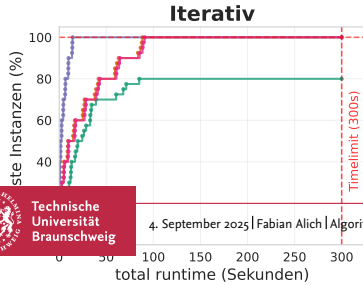
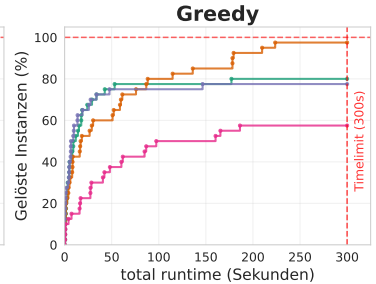
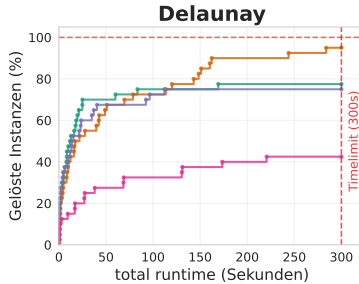
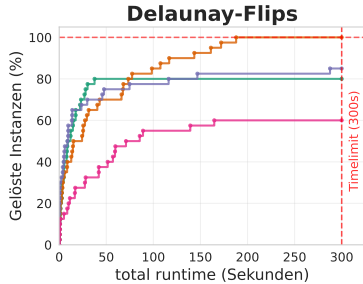
Gesamt



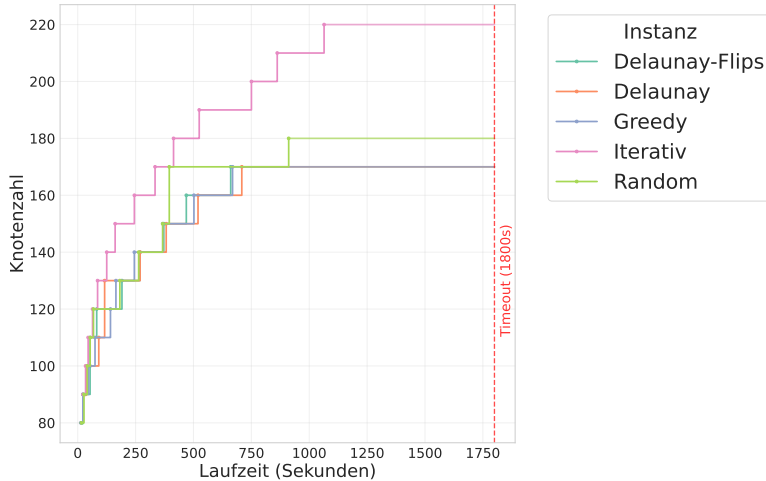
OR-Tools Bedingungen



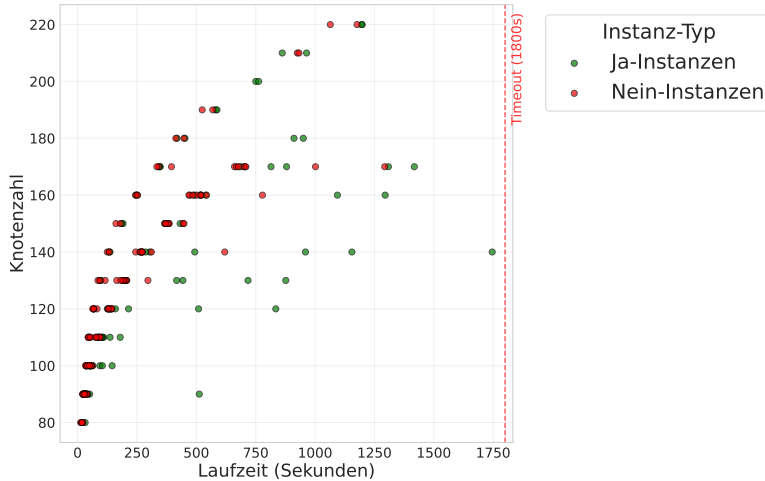
Solver Vergleich



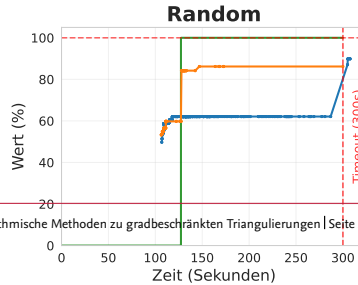
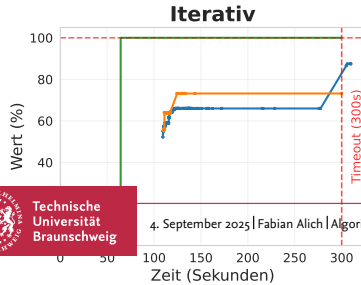
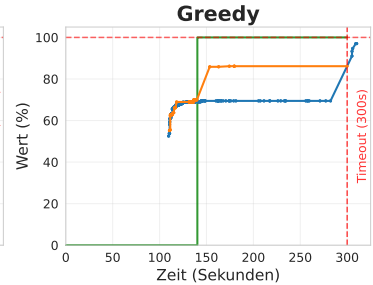
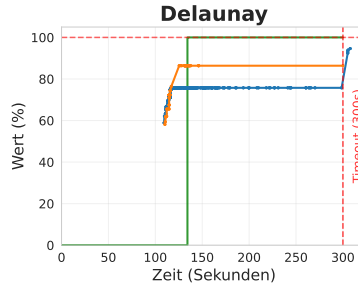
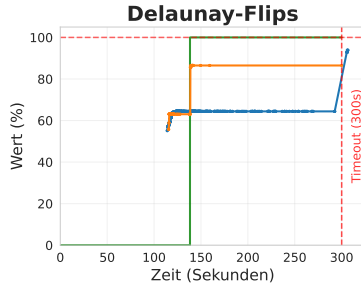
Größerer Timeout



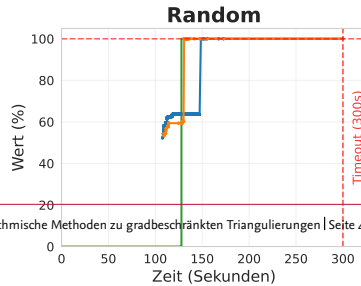
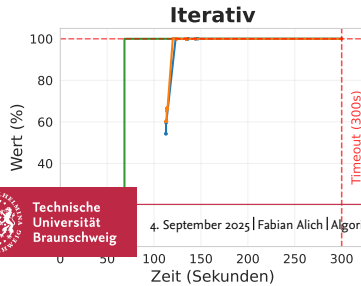
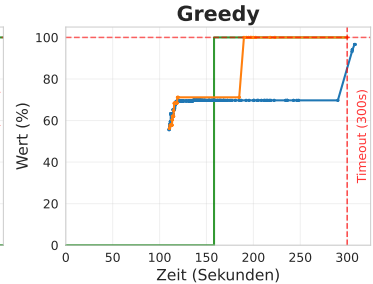
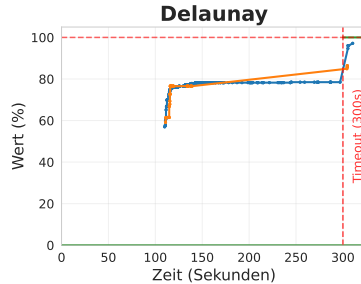
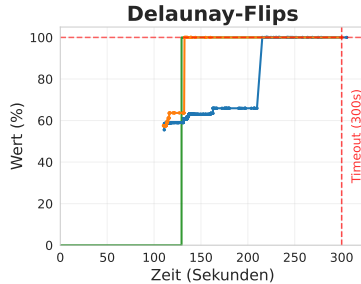
Ja Nein getrennt



Nein Instanz



Ja Instanz



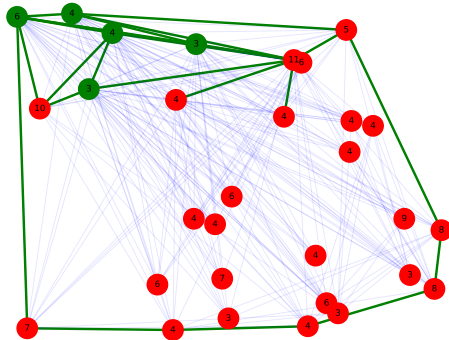
Conflict-driven clause learning

CaDiCaL Simplified Satisfiability Solver

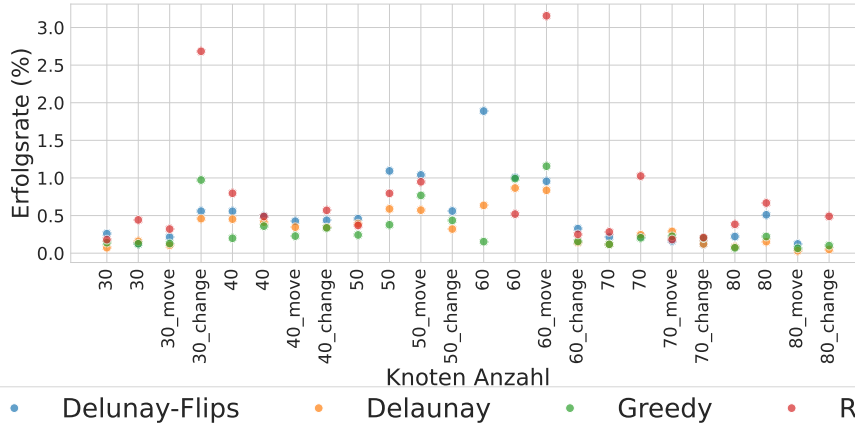
Teilinstanz

FA: Isolierte Kante Teilinstanz einfügen

Teilinstanz



Propagation



Teil IV

Fazit und Ausblick

Fazit

OR-Tools Hülle Fixieren Kanten Fixieren alternative Kantenbedingung

Fazit

FA: Bild von Instanze mit 220 Knoten

Fazit

FA: Bild von Optimierungsproblem mit 97 Prozent

Fazit

FA: Bild wo eine Propagation möglich war

Ausblick

- Skalierbarkeit
- Arbeitsspeicherverbrauch
- Dreiecksansatz Schwierigkeit Instanzen
- Optimierungen Kombinationen
- Propagation erweitern
- normalverteilte Triangulierung
- Problem abwandeln
 - Kanten vorgeben
 - Polygone